
Enhancing Neural Laplace: Robustness and Performance Analysis

A STUDY ON THE NEURAL LAPLACE MODEL

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Abstract

Neural Laplace leverages Laplace transforms to learn continuous-time representations of time-series data. This paper presents an extensive evaluation of its robustness and efficiency across a diverse set of scenarios, including chaotic systems, noisy datasets, and real-world applications. Empirical results indicate that Neural Laplace surpasses baseline models such as Neural ODE, Neural Flow Coupling, and Neural Flow ResNet in both accuracy and resilience to challenging dynamics. Notably, it achieves the lowest overall mean error (0.240355) and successfully models 26 out of 44 chaotic systems below a Mean Squared Error (MSE) threshold of 0.2, whereas other baselines meet this criterion for fewer systems. Moreover, Neural Laplace maintains lower MSE variability under noisy conditions, underscoring its capability to adapt to perturbations. On real-world datasets, it demonstrates competitive performance, particularly on the Sunspots dataset, where it attains the best MAPE (1.28) and R^2 (0.12) among the tested methods. These findings highlight Neural Laplace’s effectiveness in handling complex dynamics and call attention to areas for further improvement, including enhanced noise handling and trend modeling. Overall, this study lays a foundation for extending Neural Laplace to broader applications in dynamic system modeling and real-world data analysis.

1 Introduction

The modeling of dynamic systems is paramount in fields as diverse as engineering, finance, biology, and climate science. Accurate time-series representations in continuous-time settings are particularly critical for capturing complex behaviors and enabling reliable predictions (Chen et al., 2018; Kovachki et al., 2023; Pearl, 2009). However, traditional approaches, such as Neural ODE (Chen et al., 2018), often encounter difficulties in highly nonlinear or noisy regimes, raising concerns regarding stability and computational efficiency (Kim et al., 2021; Lehtimäki et al., 2022).

Neural Laplace models have emerged as a compelling alternative, leveraging Laplace transforms to learn continuous-time representations of time series (Cao et al., 2023; Carrel, 2024; S. Holt et al., 2023; S. I. Holt et al., 2022). By operating in the Laplace domain, these methods can address some of the limitations inherent in time-domain solvers, particularly in chaotic or stiff systems (Brown & Churchill, 2009; Gilpin, 2021). Yet, challenges such as noise sensitivity, scalability to more complex datasets, and consistency across diverse dynamic regimes remain open (Heald & Stark, 2000; Ott et al., 2023).

While early research on Neural Laplace demonstrated its potential in modeling a variety of ordinary and delay differential equations (S. Holt et al., 2023; S. I. Holt et al., 2022), many of these initial experiments focused on simplified settings. This paper extends the existing body of work to more challenging domains: chaotic systems, noisy environments, and real-world time series with intricate temporal structures. By benchmarking Neural Laplace against advanced baselines (e.g., Neural Flow (Biloš et al., 2021) and Augmented Neural ODEs (Dupont et al., 2019)), we evaluate its robustness, efficiency, and overall suitability for practical applications.

Our core objective is to uncover both strengths and limitations of Neural Laplace under demanding conditions. Specifically, we assess how it performs when faced with chaotic system trajectories (Gilpin, 2021), substantial noise (Heald & Stark, 2000), and real-world time-series data that may exhibit trends, seasonality, or non-stationary behavior (Herzen et al., 2022). In doing so, we highlight aspects of the method that excel (such as handling certain stiff or multi-scale dynamics) and pinpoint areas requiring further development (e.g., robust noise handling).

The contributions of this work are twofold. First, we offer a detailed empirical investigation of Neural Laplace¹ in scenarios ranging from chaos to real-world temporal patterns, identifying key performance metrics and bottlenecks. Second, we propose targeted enhancements—such as noise-mitigation strategies and flexible training procedures—to address these limitations, thereby paving the way for more robust continuous-time models in future research and real-world deployment.

¹<https://renkulab.io/projects/chamim/neural-laplace>

2 Background

2.1 Laplace Transforms and Continuous-Time Representations

Laplace transforms are a cornerstone in the analysis of continuous-time dynamic systems, providing a powerful method to transform differential equations into an algebraic form in the Laplace domain (Kamen & Heck, 2006; Ogata, 2020; Ziemer et al., 1998). This transformation simplifies the process of solving systems with complex dynamics, particularly those involving multiple time scales or discontinuities (Brown & Churchill, 2009). The inverse Laplace transform reconstructs the time-domain solution, making it possible to analyze and simulate system behavior. These properties have made Laplace transforms a foundational tool in control theory, signal processing, and engineering (Kuo, 1987; Oppenheim et al., 1997; Van Valkenburg, 1964).

2.2 Evolution of Neural Models for Differential Equations

Neural Ordinary Differential Equations (Neural ODEs) introduced a paradigm shift by enabling the modeling of continuous-time dynamics with neural networks (Chen et al., 2018). These models use numerical solvers to integrate time-series data, learning the underlying differential equations directly from data. However, Neural ODEs face challenges such as numerical instability when dealing with stiff equations (Kim et al., 2021) or long-term dependencies. Moreover, their reliance on time-domain integration limits their ability to model systems with abrupt changes or discontinuities. Physics-informed neural networks have further enhanced this approach by integrating physical laws into learning (Raissi et al., 2019).

2.3 Neural Laplace Models

The Neural Laplace model extends the capabilities of Neural ODEs by operating in the Laplace domain (S. I. Holt et al., 2022). Instead of directly solving differential equations in the time domain, Neural Laplace learns the Laplace transform of the system’s solution. This approach addresses several limitations of time-domain methods, such as sensitivity to noise and computational inefficiency. By leveraging the algebraic simplicity of the Laplace domain, Neural Laplace can handle a broader class of differential equations, including delay differential equations (DDEs) (S. Holt et al., 2023) and integro-differential equations (IDEs). Extensions of Neural Laplace to stochastic systems (Carrel, 2024) and applications in control (S. Holt et al., 2023) further demonstrate its versatility. Connections to neural operators (Kovachki et al., 2023) also provide a broader theoretical foundation for this method.

2.4 Prior Work and Key Limitations

Previous research has explored various enhancements to Neural ODE frameworks, such as Neural Flow models, which introduce coupling mechanisms, and ResNet-inspired architectures that improve stability (Dupont et al., 2019). Augmented Neural ODEs have addressed representational limitations by enhancing expressivity and stability (Dupont et al., 2019). Despite these advancements, challenges remain in scaling these models to real-world systems with complex and noisy dynamics (Lehtimäki et al., 2022). The application of Neural ODEs to circuit modeling has revealed additional capabilities and inherent challenges (Xiong et al., 2022). Efforts to address these issues include tuning strategies for stability and faster convergence (Akhtar, 2023) and adaptive control mechanisms (Chalvidal et al., 2020).

The Neural Laplace model represents a significant step forward by providing a unified framework capable of handling diverse dynamic behaviors (S. I. Holt et al., 2022). Extensions such as the Laplace Neural Operator (Cao et al., 2023) leverage the Laplace transform to improve generalization and interpretability in differential equation modeling. However, its application to chaotic systems, noisy datasets, and real-world time series has yet to be comprehensively evaluated. Moreover, the integration of stochastic dynamics (Carrel, 2024) and delayed system control (S. Holt et al., 2023) represent promising areas for further exploration.

3 Model

The Neural Laplace model, introduced by (S. I. Holt et al., 2022), is a unified framework designed to learn a broad spectrum of differential equations (DEs) by operating in the Laplace domain. This approach addresses limitations found in traditional Neural Ordinary Differential Equations (Neural ODEs) (Chen et al., 2018), particularly their challenges in modeling systems with long-range dependencies, discontinuities, and numerical instabilities when dealing with stiff ODEs or piecewise forcing functions (Kim et al., 2021).

3.1 Model Architecture

Initial Condition Encoder

The model begins with an encoder network, which processes observed data points to produce a representation of the initial condition. This representation captures the system’s state at the starting point and is crucial for accurate trajectory prediction.

The encoder is implemented as a multi-layer perceptron (MLP) that maps the input data to a fixed-dimensional latent space. To handle diverse input formats, such as irregular time series or multidimensional signals, the encoder incorporates feature extraction layers, including positional encodings for temporal inputs. These layers ensure that the initial condition embedding preserves critical information about the system’s starting state, even for complex or noisy datasets.

Laplace Representation Network

Instead of modeling dynamics directly in the time domain, Neural Laplace operates in the Laplace domain. It employs a neural network to approximate the Laplace transform of the system’s solution (S. I. Holt et al., 2022). This allows the model to handle complex history dependencies and discontinuities more effectively, as these features become more tractable in the Laplace domain.

The Laplace representation network is designed to approximate the solution’s Laplace transform $F(s)$ efficiently and accurately. It uses a parameterized function $g(\cdot; \theta)$, where θ represents the learnable parameters of the network. The network takes the Laplace-domain variable s as input, along with the initial condition embedding produced by the encoder, and outputs an approximation of $F(s)$. To ensure stability and accuracy, the Laplace representation network incorporates techniques such as spectral normalization and weight regularization (Dupont et al., 2019). These techniques constrain the network’s outputs, preventing divergence or overfitting in the complex Laplace domain.

Inverse Laplace Transform

To convert the solution back to the time domain, the model utilizes an ILT algorithm. This step reconstructs the time-domain solution from its Laplace representation, en-

abling the generation of continuous-time trajectories (Chen et al., 2018).

The ILT step is implemented using numerical techniques such as Fourier series expansions and adaptive quadrature methods. These methods approximate the Bromwich integral, ensuring accurate reconstruction of $x(t)$ from $F(s)$ over a wide range of frequencies. The model also supports batched and parallelized ILT computations, making it scalable for large datasets or real-time applications. Additionally, the ILT module includes regularization mechanisms to suppress artifacts introduced by numerical errors, such as Gibbs oscillations or truncation effects.

By combining these components, the Neural Laplace model provides a robust framework for learning differential equation dynamics in the Laplace domain. The architecture’s modularity enables it to be extended or adapted for specialized tasks, such as hybrid time-frequency analysis or solving partial differential equations (PDEs).

3.2 Mathematical Foundation

The mathematical foundation of Neural Laplace is rooted in the properties of the Laplace transform, which enables the representation of a wide class of DEs in a compact and efficient manner. The Laplace transform of a time-domain signal $x(t)$ is defined as:

$$F(s) = \mathcal{L}\{x(t)\}(s) = \int_0^\infty e^{-st} x(t) dt, \quad (3.1)$$

where s is a complex number (Oppenheim et al., 1997). This transformation maps the time-domain trajectory to a complex domain, where operations such as differentiation and integration become algebraic. For example:

$$\mathcal{L}\{x'(t)\}(s) = sF(s) - x(0). \quad (3.2)$$

Using the Laplace transform, Neural Laplace can model diverse DEs, including delay differential equations (DDEs), integro-differential equations (IDEs), and stiff ODEs. The inverse Laplace transform (ILT) is used to reconstruct the time-domain solution $x(t)$ from its Laplace representation $F(s)$:

$$x(t) = \mathcal{L}^{-1}\{F(s)\}(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} F(s) e^{st} ds, \quad (3.3)$$

where σ is chosen such that all singularities of $F(s)$ lie to the left of the contour (Brown & Churchill, 2009).

The ILT forms a critical component of Neural Laplace as it reconstructs time-domain solutions from their Laplace representations. Computationally, the ILT is evaluated using numerical techniques such as Fourier series expansion (Oppenheim et al., 1997), which approximates the Bromwich contour integral by summing discrete frequency components. This approach efficiently computes $x(t)$ at arbitrary time points, allowing the model to handle stiff systems and discontinuities. Unlike traditional methods, which rely on numerical solvers with stepwise integration, the ILT offers global reconstructions of solutions, improving numerical stability and generalization.

A unique aspect of Neural Laplace is its use of stereographic projection to map the complex Laplace domain onto a bounded representation on the Riemann sphere. This mapping eliminates singularities and improves the stability and generalization of the

learning process. The stereographic projection maps a point $s \in \mathbb{C}$ onto the surface of a sphere by:

$$(\theta, \phi) = u(s) = \left(\arctan \left(\frac{\text{Im}(s)}{\text{Re}(s)} \right), \arcsin \left(\frac{|s|^2 - 1}{|s|^2 + 1} \right) \right), \quad (3.4)$$

where θ and ϕ are angular coordinates. The inverse transform v maps these coordinates back to the complex plane:

$$s = v(\theta, \phi) = \tan \left(\frac{\phi}{2} + \frac{\pi}{4} \right) e^{i\theta}. \quad (3.5)$$

This mapping ensures that the point at infinity in the complex domain corresponds to the north pole of the sphere, while the rest of the domain is compactly represented. By bounding the Laplace representation in this way, the model avoids issues with singularities and enhances numerical stability during learning (Brown & Churchill, 2009).

3.3 Adapting the Model to Real-World Datasets

The Neural Laplace model is implemented to take the first half of a time series as input and predict the second half. A key requirement for this process is that the time points within the series must be unique. This constraint means that the model cannot be directly trained to predict overlapping time intervals, such as time points (3,4) and (5,6), without additional preprocessing (Herzen et al., 2022).

To address this, we employed a repetition-based forecasting strategy. The original time series is divided into smaller chunks of a fixed size. Within each chunk, equally distributed time points between 1 and the chunk size are selected for training. The model predicts the second half of the chunk step by step until the entire series is forecasted. This approach allows the model to handle overlapping time intervals while ensuring robust training and testing on diverse temporal patterns.

By chunking the time series, we enhance the model’s capacity to generalize to various temporal resolutions and improve its ability to handle datasets with unevenly spaced time points. This preprocessing step ensures that the Neural Laplace model remains effective for real-world applications, where time series data often exhibit non-uniform sampling and overlapping temporal dependencies.

4 Results

The experiments conducted in this study involved evaluating the Neural Laplace model against multiple baseline models on different datasets and systems.

4.1 Experiments on Chaotic Systems

Neural Laplace was tested on chaotic systems from the Dysts dataset. Baseline models included Neural ODE, Neural Flow Coupling, and Neural Flow ResNet. Although experiments were also conducted using N-BEATS, TCN, and RNN models, these methods failed to capture even the simplest system dynamics. Consequently, the detailed results from these trials are omitted from this report.

4.1.1 Performance of Neural Laplace

The performance of Neural Laplace on chaotic systems from the Dysts dataset is discussed in detail below. Neural Laplace demonstrates significant advantages in terms of accuracy and robustness when compared to the baseline models.

Grouped Statistics Across Files

The overall mean performance of each method across files is summarized in Table 4.1. This table was generated by taking the mean of experiments conducted on 44 chaotic systems from the Dysts dataset.

Method	Mean (Overall)
Neural Laplace	0.240355
NODE (Euler)	0.312405
Neural Flow Coupling	0.544674
Neural Flow ResNet	0.407900

Table 4.1: Grouped Statistics (Overall Mean Across Files, Derived from 44 Chaotic Systems in the Dysts Dataset)

Number of systems accurately modeled by Neural Laplace

The number of systems achieving a Mean Squared Error (MSE) below 0.2 for each method is shown in Table 4.2. These results highlight the robustness of Neural Laplace in comparison to other baseline models.

Cumulative Distribution of MSE Values

The cumulative distribution of Mean Squared Error (MSE) values for all methods is depicted in Figure 4.1. This plot illustrates the sorted MSE values across the 44 chaotic

Method	Count of Systems with MSE < 0.2
Neural Laplace	26
NODE (Euler)	21
Neural Flow Coupling	9
Neural Flow ResNet	11

Table 4.2: Number of Systems with MSE Below 0.2 (Derived from 44 Chaotic Systems in the Dysts Dataset)

systems from the Dysts dataset, highlighting the comparative performance of the models.

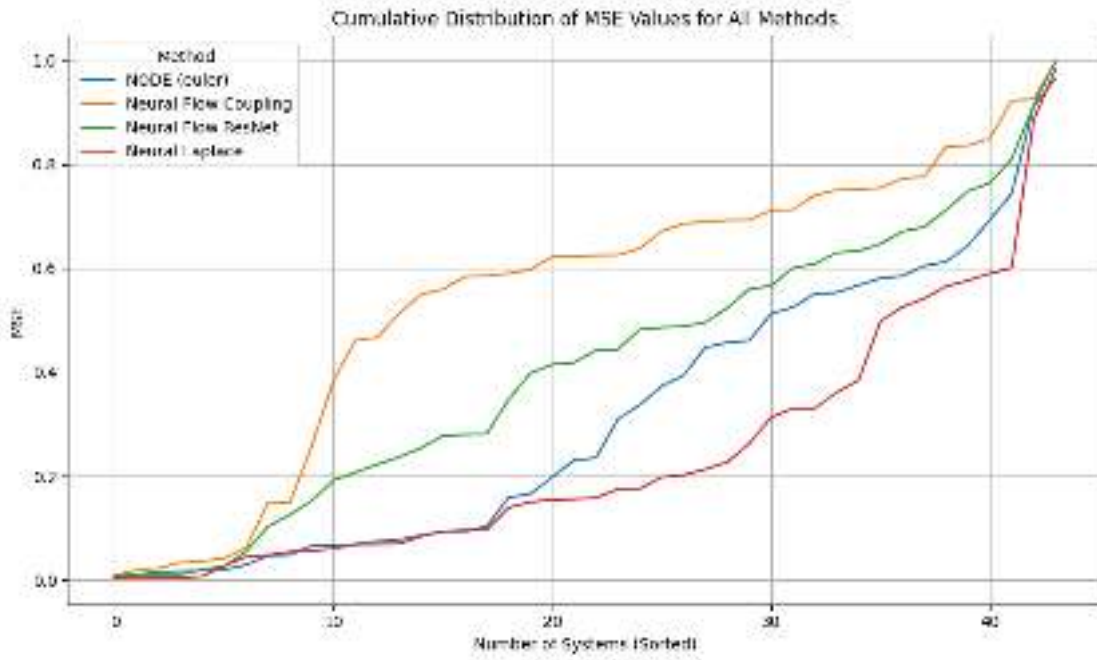


Figure 4.1: Cumulative Distribution of MSE Values for All Methods. Neural Laplace exhibits consistently lower MSE values compared to other methods, particularly in the lower MSE range.

Analysis The cumulative distribution plot in Figure 4.1 provides insights into the performance variability across the chaotic systems. Neural Laplace demonstrates superior performance with the lowest cumulative MSE values across the majority of systems, emphasizing its robustness. In comparison, NODE (Euler) achieves slightly higher MSE values than Neural Laplace but still outperforms other baseline models. In contrast, Neural Flow ResNet and Neural Flow Coupling exhibit higher cumulative MSE values, indicating lower performance consistency relative to Neural Laplace and NODE (Euler).

Correlation between MSE and Lyapunov Exponent

The relationship between the Mean Squared Error (MSE) of Neural Laplace and the Lyapunov exponent of chaotic systems is shown in Figure 4.2. This scatter plot highlights the impact of the Lyapunov exponent on model performance.

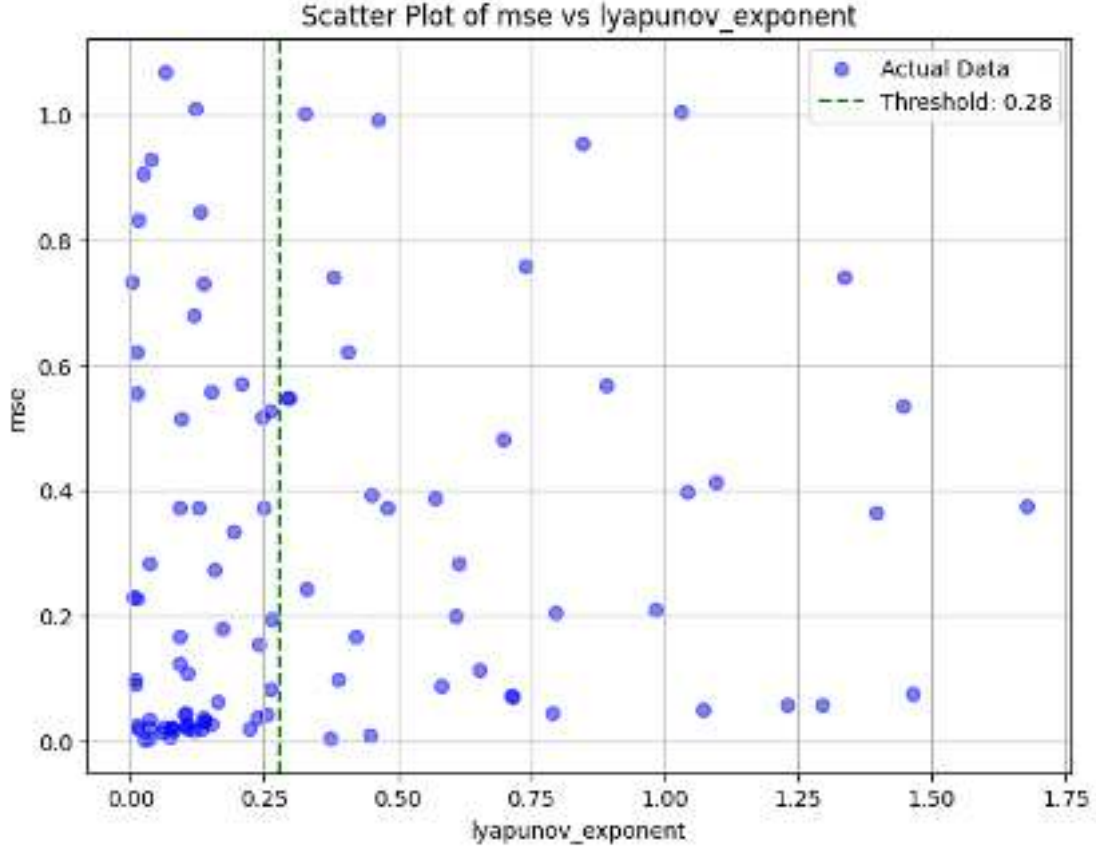


Figure 4.2: Scatter Plot of MSE vs Lyapunov Exponent for Neural Laplace on Chaotic Systems from the Dysts Dataset. The vertical dashed line represents a Lyapunov exponent threshold of 0.28. Systems with lower Lyapunov exponents tend to have lower MSE values.

Analysis The scatter plot in Figure 4.2 reveals several insights. Systems with a Lyapunov exponent below 0.28 generally exhibit lower MSE values, suggesting that Neural Laplace performs better on systems with less chaotic behavior. Conversely, for systems with higher Lyapunov exponents, the MSE values vary widely, indicating an increased difficulty in accurately modeling highly chaotic systems.

4.2 Experiments on Noisy Chaotic Systems

Neural Laplace was further tested on noisy chaotic systems from the Dysts dataset. The evaluation metrics and comparative performance against baseline models remain consistent with the previous experiments.

4.2.1 MSE Distribution With and Without Noise

The distribution of Mean Squared Error (MSE) values for Neural Laplace on chaotic systems, both with and without added noise, is shown in Figure 4.3. This histogram provides insights into how noise impacts the model’s performance.

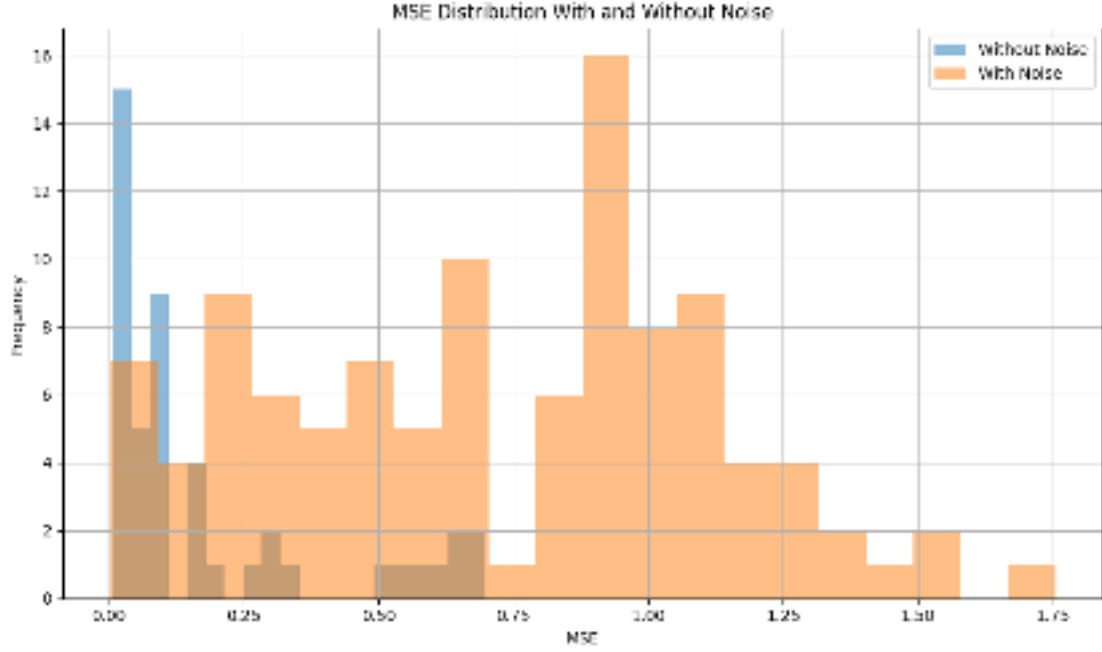


Figure 4.3: MSE Distribution With and Without Noise. The blue bars represent MSE values without noise, and the orange bars represent MSE values with noise. Neural Laplace demonstrates lower MSE values without noise, while noise increases the variability and average MSE.

Analysis The histogram in Figure 4.3 highlights several observations. Without noise, the MSE values are clustered around lower figures, indicating better performance and less variability in model predictions. In contrast, the addition of noise increases both the average MSE and its variability, as evidenced by the broader distribution of values.

4.2.2 Comparison of Models Below MSE Threshold (0.5)

The comparison of the number of models achieving a Mean Squared Error (MSE) below 0.5, with and without added noise, is shown in Figure 4.4. This plot demonstrates the effect of noise on the proportion of models maintaining low MSE values.

Why 0.5 as the Threshold? The threshold of 0.5 was chosen for several reasons. Through visual observation, it was noted that when the MSE is below 0.5, the model generally captures the underlying dynamics of the system, signifying effective learning. Additionally, due to the added noise, Neural Laplace faces challenges in learning the noisy patterns, as it is a continuous method designed to capture smooth representations. Setting a higher threshold accounts for the increased complexity of understanding the system under noisy conditions. Ultimately, this threshold serves as an indicator that the model has attained a basic understanding of the system's dynamics, even in the presence of noise.

Analysis The bar chart in Figure 4.4 shows that without noise, a larger number of models achieve MSE values below the 0.5 threshold, reflecting better overall performance and a more effective learning of system dynamics. In contrast, with noise, the number

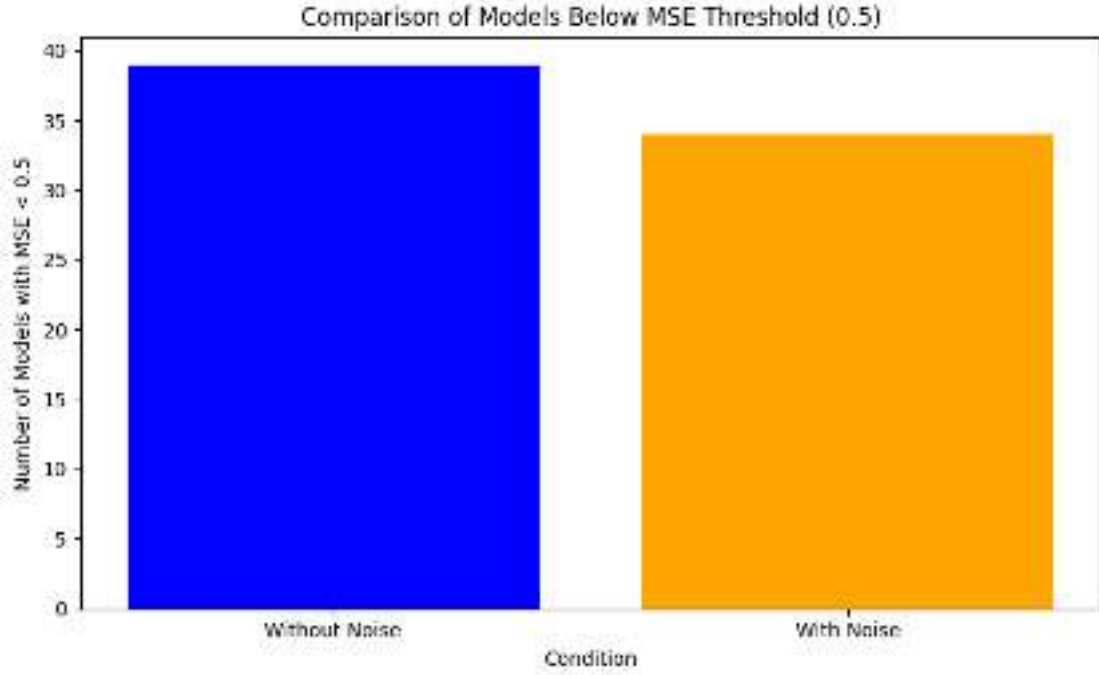


Figure 4.4: Comparison of Models Below MSE Threshold (0.5). The blue bar represents the number of models without noise achieving an MSE below 0.5, while the orange bar represents the same metric with noise.

of models falling below this threshold decreases, demonstrating the increased difficulty Neural Laplace faces when learning in noisy conditions.

4.3 Experiments on Real-World Data

Neural Laplace (NL) was evaluated on three real-world datasets from the Darts library: the AirPassengers dataset, the Monthly Milk dataset, and the Sunspots dataset. The performance of NL was compared against NBEATS, TCNModel, and RNNModel using the Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), Coefficient of Determination (R^2), and other metrics where available.

4.3.1 AirPassengers Dataset

Table 4.3 summarizes the performance of Neural Laplace and baseline models on the AirPassengers dataset. This dataset captures the number of monthly airline passengers in thousands from 1949 to 1960.

Analysis While Neural Laplace achieves competitive MAPE values, it is outperformed in MSE and R^2 metrics by the other models. It fails to accurately model the upward trend of the dataset.

Model	MAPE	MSE	R ²	Training Time	# Parameters
Neural Laplace (NL)	0.15	6,197.04	-0.02	1m55s	51,000
NBEATS	0.07	1,446.30	0.76	2m0s	6.2M
TCNModel	0.05	637.16	0.92	1m18s	152
RNNModel	0.09	2,529.66	0.58	2m9.8s	726

Table 4.3: Performance of Neural Laplace and Baseline Models on the AirPassengers Dataset.

4.3.2 Monthly Milk Dataset

Table 4.4 presents the performance of the models on the Monthly Milk dataset, which records monthly milk production per cow in pounds from 1962 to 1975.

Model	MAPE	R ²
Neural Laplace (NL)	0.36	0.38
NBEATS	0.06	-0.10
TCNModel	0.04	0.45
RNNModel	0.05	0.21

Table 4.4: Performance of Neural Laplace and Baseline Models on the Monthly Milk Dataset.

Analysis On the Monthly Milk dataset, TCNModel demonstrates the best overall performance in terms of MAPE and R². Neural Laplace achieves reasonable R² values but lags in MAPE compared to the other models.

4.3.3 Sunspots Dataset

Table 4.5 shows the performance metrics for the Sunspots dataset, which contains monthly mean sunspot numbers from 1749 to 1983.

Model	MAPE	R ²
Neural Laplace (NL)	1.28	0.12
NBEATS	2.47	-0.74
TCNModel	1.38	-0.13
RNNModel	2.82	-1.04

Table 4.5: Performance of Neural Laplace and Baseline Models on the Sunspots Dataset.

Analysis On the Sunspots dataset, Neural Laplace achieves the best MAPE and R² among all models, outperforming NBEATS, TCNModel, and RNNModel. Despite the challenges posed by the dataset, Neural Laplace demonstrates its ability to generalize better than other methods in this case.

5 Discussion and Conclusion

5.1 Summary of Results

Neural Laplace consistently outperformed baseline models on chaotic systems from the Dysts dataset. Its overall mean MSE was lower than Neural ODE (Euler), Neural Flow Coupling, and Neural Flow ResNet, demonstrating robustness and accuracy across 44 chaotic systems. In particular, Neural Laplace achieved better performance for systems with lower Lyapunov exponents, indicating sensitivity to the degree of chaos in the underlying dynamics.

When noise was introduced, Neural Laplace’s performance declined, as seen in broader MSE distributions and fewer systems meeting lower MSE thresholds. This underscores its sensitivity to noisy inputs and the need for enhanced noise-mitigation strategies.

For real-world datasets (AirPassengers, Monthly Milk, Sunspots), Neural Laplace showed mixed results. It performed best on the Sunspots dataset, capturing quasi-periodic fluctuations more effectively than the baselines. However, it struggled on datasets with strong trends and seasonality, such as AirPassengers and Monthly Milk, highlighting areas where classical time-series models (e.g., TCN, RNN, or NBEATS) can outperform it.

5.2 Advantages and Limitations

By learning in the Laplace domain, Neural Laplace exhibits wide applicability, handling stiff, delay, and integro-differential equations more naturally than time-domain methods. The use of the Laplace transform simplifies discontinuities and multi-scale behavior, thereby mitigating numerical instabilities that are common in Neural ODE approaches. Furthermore, Neural Laplace demonstrates strong performance on moderately chaotic systems, consistently achieving lower MSE than its competitors and showing promise for tasks involving complex dynamical phenomena.

However, Neural Laplace also presents several limitations. Its performance drops significantly in noisy conditions, suggesting a need for dedicated noise-handling or denoising modules. The method underperforms on datasets with pronounced trends or strong seasonality, indicating potential challenges in modeling long-range, non-stationary patterns. Additionally, while Laplace-domain modeling offers benefits, the inverse Laplace transform introduces computational overhead relative to direct time-domain methods. Finally, restricting the model to fixed time grids during training can reduce its flexibility, especially when dealing with unevenly sampled or highly variable datasets.

5.3 Conclusion

In summary, Neural Laplace advances continuous-time modeling by learning solutions in the Laplace domain. It demonstrates notable advantages on chaotic and certain real-world tasks, outperforming standard Neural ODEs in handling moderate chaos and

complex dynamics. However, limitations involving noise robustness and strong trend modeling must be addressed for broader applicability.

Future work could explore a number of promising directions. One avenue involves developing noise-robust extensions, such as incorporating denoising or probabilistic approaches to better handle uncertainties and measurement noise. Another direction is enhancing adaptability to trends by combining Neural Laplace with time-domain architectures, aiming to more effectively capture non-stationarities and seasonal components.

Overall, Neural Laplace stands as a promising framework for modeling continuous-time dynamics, particularly chaotic systems, while ongoing refinements aim to bolster its resilience to noise and strong trends. This balance between theoretical generality and practical performance paves the way for future developments in data-driven differential equation modeling.

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